

can be processed using low-cost equipment such as ink-jet printers or coating equipment to make photographic film, which reduces both capital and manufacturing costs compared to conventional solar-cell manufacturing.

Some organic materials

tris(8-hydroxyquinolino)aluminium (Alq3). This organic material is most commonly used for the light-emitting layer of OLEDs.

Oligothiophene derivatives. Oligothiophene derivatives bear the styryl (DS-*n*T) or pyrene (Py-*n*T) unit as terminal substituent. Thin-film field-effect transistors incorporating DS-*n*T show high electrical performance, such as mobilities as high as $0.1 \text{ cm}^2/\text{Vs}$, along with exceptional stability under ambient conditions. Especially, the longer oligomer DS-4T containing the quaterthiophene core gives rise to devices that show no decrease in performance after more than 17 months of storage and under continuous operation.

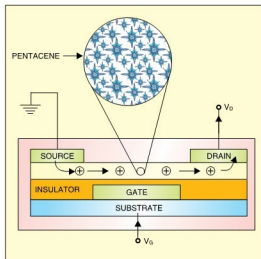
6,7,15,16-tetrakis(alkylthio)quinoxalino[2,3':9,10]-phenanthro[4,5-abc]phenazine, TQPP-[SR]4. These materials have been synthesised for $\text{R}=\text{C}_{12}\text{H}_{25}$ and C_6H_{13} as solution-processable potential charge transport materials for organic electronics applications.

bis(tri-isopropylsilyl)ethynyl)pentacene (TIPS-pentacene). Using this material, solution-processed OTFTs with mobility of about $1.5 \text{ cm}^2/\text{Vs}$ are fabricated. This is the highest mobility reported to date for solution-processed OTFTs.

Related technologies

MEMS. Polymer-based MEMS are rapidly gaining momentum due to their potential for conformability and other special characteristics not available with silicon microsystems. Polymer-based nano- and micro-devices are flexible, chemically and biologically compatible, and available in many varieties. These can also be fabricated in truly 3-D shapes.

Nanoimprint lithography. It is



Structure of an organic semiconductor p-type thin-film transistor with top contacts

Comparison between Organic and Silicon Electronics

Features	Organic electronics	Silicon
Cost	\$55.5/m ² (\$5/ft ²)	\$1111/m ² (\$10Q/ft ²)
Fabrication cost	Low capital	\$1-10 billion
Device size	3m (10 ft) × roll to roll	<1 m ²
Material	Flexible plastic substrate	Rigid glass or metal
Required condition	Ambient processing	Ultra-clean room
Process	Continuous direct printing	Multistep photolithography

an emerging technology for high-throughput nanoscale patterning. Nano-imprinting uses a hard mould to mechanically deform a polymer material in order to create nanoscale structures, which provides unprecedented resolution and speed in nanopatterning. Due to their compatibility with polymer materials, nanoimprint and its variants find unique applications in polymer-based devices, including organic electronics and photonics.

Future of organic electronics

Organic electronics has opened the door for various advancements in technology. Carbon nanotube based transistors will replace Si- and Ge-based transistors in the coming future. Due to its scalability, a large number of devices can be integrated on a single chip. Organic electronic

devices will provide an effective solution to pollution by minimising e-waste due to their bio-degradability. Some interesting applications of organic electronics are mentioned below.

Smart textiles. Electronic devices can be integrated into textiles to monitor heart-rate and other vital signs, control embedded devices (MP3 players), etc.

Lab on a chip. It's a device that incorporates multiple laboratory functions in a single chip. As the name suggests, it actually enables

the functionality of a lab on a miniature device. You can take samples of the substance/component to be tested, sometimes at a remote location or the patient's bedside in the case of medical diagnostics, and analyse it right then and there.

A lab-on-a-chip can be used in environmental testing to determine, for example, whether there is contaminated water or harmful chemicals in the air. These devices could have multiple applications in medical diagnostics too.

Portable compact screens. Micro-displays have the potential to create new opportunities for ultra-portable digital appliances and entertainment products. All that remains to be done is fine-tuning of silicon circuitry designs for the best performance and mechanical integration of colour filters for full-colour capability. ●

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